Aravind Kannappan

San Francisco Bay Area, California
1-408-591-5449
aravinds.kannappan@gmail.com
LinkedIn | GitHub | Website
Open To Relocation
US Citizen

EDUCATION

New York University
MS in Applied Statistics
Specialization in Machine Learning
Aug 2024 – Expected Aug 2026

Baylor University
BS in Statistics
Minor in Biology
Aug 2020 – Aug 2024

TECHNICAL SKILLS

Languages:
Python, TypeScript, SQL, R, C++
AI/ML Frameworks:
LangChain, LangGraph, PyTorch, Hugging Face, vLLM, Scikit-learn, RAG, Prompt Engineering, Tensorflow, MCP
Frontend: React, REST APIs
Cloud & DevOps:
AWS (S3, EC2, Bedrock), GCP, Docker, Kubernetes, CI/CD, Git

COURSEWORK

- NLP, Deep Learning, Data Engineering, Distributed Systems, Big Data, Bayesian Methods, Biostatistics
- Data Structures & Algorithms, Linear Algebra, Calculus, Database Systems, System Design, Operating Systems

Technical Writing BLOG

- Wrote posts on optimization, generalization, and probabilistic modeling from first principles.
- Built reproducible Python notebooks, training scripts, and benchmarks.
- Studied failure modes (leakage, shift, calibration) and proposed fixes.

Publications

- **Multiple Myeloma Gene Signatures (The Oncologist, 2024)** Built statistical models in Python for mutation, pathway, and survival analysis, identifying gene signatures. Link
- **Agentic AI Systems for Clinical Reasoning (Book Chapter, 2026)** Developed frameworks for integrating clinical & social data to improve decision-making and patient-centered care. Link

CERTIFICATIONS

Google Cloud Professional MLE, AWS MLE & Solutions Architect Associates, NVIDIA Associate: Gen AI & LLMs

SUMMARY

Two time founder with production experience designing LLM powered systems, RAG pipelines, and multi agent orchestration across diverse domains. Published researcher and open source contributor advancing work in AI safety and evaluation. Strong foundation in Machine Learning Systems and software engineering, working across the full stack in fast moving, ambiguous environments

WORK EXPERIENCE

Research Engineer Fellow — Jun 2026 – Present
Eleuther AI | Summer of Open AI Research — *Remote*

- Applied SAEs and linear probes on residual-stream activations across 24 transformer layers to localize galaxy-morphology emergence, reaching 0.91 probe accuracy using Pytorch
- Computed mutual k-NN alignment (k=10) across 4 foundation models on 120K astronomical samples to test Platonic Representation Hypothesis via cosine similarity and rank correlation
- Built activation patching and feature ablation for causal interventions over 16K-feature SAE dictionaries; SLURM multi-GPU pipeline (8 GPUs) cut large-scale feature extraction 6x
- Trained crosscoders across 4 models to isolate shared galaxy-decomposition features; validated inter-model agreement at 0.89 F1 on held-out probing classifiers
- Built layer-wise FAISS-indexed feature-extraction pipeline over 2M galaxy embeddings; KL-divergence shift detection pinpointed representation-emergence thresholds across 24 layers

Cofounder, Software Engineer AI/ML Website | GitHub — Sep 2025 – May 2026
Replays AI | AI-Powered Immersive Post-Game Recaps — *New York, NY*

- Designed ingestion pipeline writing 5M+ structured play-by-play events from the ESPN API into PostgreSQL as system of record; stored raw video referenced by pointer, cutting compute by 40%.
- Enforced schema validation & player disambiguation across 50K+ records in Python, cutting 95% duplicate entries; anchored video segmentation & CV inference for accurate highlight alignment
- Orchestrated 3 parallel agentic inference pipelines via asyncio spanning event feature extraction, Claude Vision play classification (14 labels, confidence-scored), and reducing inference by 35%.
- Decoupled storage and inference from the frontend via 12+ FastAPI REST endpoints; implemented Redis recap caching, sustaining sub-second responses on cache hits at 85% hit rate
- Built React 18 and TypeScript frontend with Vite and TanStack Query rendering personalized game feeds and fan-perspective AI recaps, reducing load time by 28% via component memoization.

Founder and ML Engineer Website | GitHub — Jul 2023 – Jul 2025
Synthure | Mulit-Agent Clinical Intelligence — *New York, NY*

- Built hybrid retrieval over 1.43M ICD-10 embeddings (pgvector cosine + BM25 fallback, IVFFlat-indexed) for clinical code grounding, reaching MRR@5 of 0.91 on offline benchmarks
- Architected multi-agents communicating through typed IR data classes, confidence-routing high-risk cases (score ≥50) to Claude Sonnet and the rest to Haiku with an async dispatcher
- Trained a Gradient Boosting denial predictor on 38,924 clinical transcriptions (AUC-ROC 0.87) to triage claims and gate LLM escalation upstream of generation
- Enforced 3-tier autonomy guardrails in code with forced tool_use schemas and post-generation citation validation, eliminating fabricated facts (2.3% drift) across 60K+ records
- Cascaded NER stack at 94.2% on held-out MTSamples, served through 42 endpoints across 4 portals; deployed on Supabase with JWT auth and HIPAA PHI audit logging

OPEN SOURCE

PodBench Website | Github
LLM-agent benchmark on deterministic, resettable SQL environments; KEDA-autoscaled K8s (2–24 replicas), per-run token metering, ∼10x cheaper via prompt caching, cost-vs-reward frontier models

Popper Website | Github
Breaks faulty proof specs via counterexamples pre-verification, cutting proof-search compute ∼20%; 176/178 unfaithful specs caught (99% recall, 0 false alarms, F1 0.99)

Context Forge Website | Github
Pareto benchmark for LLM context compression (GPT-4o/4/2, SQuAD/GovReport): ∼92% quality at ∼51% fewer tokens vs. 71% truncation; droppability classifier at 0.96 ROC-AUC

VeriGrad RL Website | Github
Mechanistic-interpretability framework for safety RL: policy learns verifier-scored activation interventions; With OOD jailbreak/over-refusal evals (0% jailbreak, 0% over-refusal)

PROJECTS

MOSAIC Website | Github
Bayesian disease-surveillance system across 39 countries; validated on 1,334 forecasts (AUROC 0.917)

Covera Website | Github
Insurance engine; MEPS Monte-Carlo simulation over 1,600 CMS plans, rank by risk-adjusted cost

RobustSight: Advancing AI Safety and Alignment Github
CV AI-safety framework; saliency training raised PGD robustness & IoU to 45% at under 3% loss